DHEERAJ PANEER SELVAM

+1 609-436-0149

Senior Python Developer

dhep5791@gmail.com

Professional Summary:

- Around 5+ **years** of IT Experience and Research in **designing, developing, testing,** and **implementing various** stand-alone and **client-server architecture**-based enterprise application software in **Python** on different domains.
- Skilled experience in **Python** with proven expertise in using new tools and technical developments (libraries used: libraries- **NumPy, SciPy, matplotlib, Pickle, Pandas, RLlib, Pandas, networks, Pytorch, MySQL** dB for database connectivity)
- Machine Learning experience in **LLMs, Ranking, Retrieval, Computer vision and Reinforcement Learning**
- Research Awards and **SOTA publications.**
- Good experience in developing **web applications** implementing Model View Control architecture using **Django, Flask web application frameworks**.
- Experience in AWS cloud platform as worked directly on the AWS development team at amazon like **AWS Glue, AWS Lambda, AWS Appflow, AWS IAM, AWS Event Bridge, AWS Cloud formation, AWS Stack** and many other services
- Good knowledge on front end frameworks like CSS Bootstrap.
- Experience with Design, code, and debug operations, reporting, data analysis and web applications utilizing Python.
- Excellent experience in Python development under Linux OS (Debian, Ubuntu, RedHat Linux)
- Used all major ETL transformations to load the tables through Informatica mappings.
- Sound experience in **Object Oriented Programming** using concepts like **Multi-Threading**, Exception Handling and Collections.
- Developed a **GraphQL** backend **Node.js** app and deployed it **on AWS Lambda** via Apex Up.
- Everything running on Linux and making use of **Google cloud SDKs**.
- Knowledge about setting up **Python REST API framework** using **Django**.
- Experience in working with **Python ORM Libraries** including **Django ORM**.
- Good Knowledge in implementation of **Python best Practices (PEP-8).**
- Good at writing SQL Queries, Stored procedures, functions, packages, tables, views, triggers using relational databases like MySQL, Oracle, DB2.
- Proficient in using **editors Eclipse, PyCharm, PyScripter, Notepad++** and Sublime Text while developing different applications.
- Experienced in developing test automation framework with Python scripting& Selenium.
- Having good knowledge in using NoSQL databases like Kubernetes and Mongo DB (2.6, 2.4).
- Having good knowledge of USB **network testing and interfacing.**
- Worked on various operating systems like **Windows** and **LINUX**.
- Experienced in Shell Scripting.
- Communication skills, efficient time management and **organization skills**, ability to handle multiple tasks and work well in a team environment.

Technical Skills:

Programming Languages	Python, Java, JavaScript, SQL and Shell Scripting
Python Libraries	Pytorch, Tensorflow, Flask, Django, httplib2, RLlib, HTML/CSS, Bootstrap, NumPy, Matplotlib, FastAPI, Pickle, Scikit-learn, SciPy, PyTables, Pandas, Tensorflow.

Frameworks	JIRA, Django, web2py, MongoDB, Hadoop/ Big Data, RASA, Spark
Technologies	React, Java Script, JQuery, XML, AngularJS, Version Control GIT (GitHub), Node.js,
Cloud Environment	AWS Services, AWS Glue, AWS Lambda, AWS Appflow, AWS IAM, AWS Event Bridge, AWS Cloud formation, AWS Stack
Test Build and Integration tools	AWS, Microsoft Azure, IBM Watson, Ansible
Databases	MySQL, Oracle, SQL Server
Operating System	Windows, Linux
MS Office Tools	Word, Power Point, Excel, Outlook

Professional Experience:

Client: Amazon (AWS)/Intellilink Technologies, Bellevue, WA (Remote) Oct 2023 – Present
Role: Senior Python Developer

Responsibilities:

- **Accomplished a 30% increase in data transfer efficiency** and a 40% reduction in processing time for **machine learning model data pipelines**, by optimizing ETL processes using **AWS Glue** and **AppFlow**, ensuring faster and reliable data flow for training and inference tasks in ML applications.
- **Led the automation of CI/CD pipelines using AWS CodePipeline and CodeCommit**, accelerating **model deployment cycles by 50%** and reducing deployment errors by 20%, specifically for deploying machine learning models, improving overall model reliability in production environments.
- **Architected scalable microservices and database solutions on AWS** using Java and Python, as measured by a 25% reduction in query execution time, optimizing data storage and retrieval for **ML model training and real-time predictions**.
- **Reduced system downtime by 35%** by improving incident resolution times and optimizing **ML model troubleshooting** through AWS tools like **Lambda** and **Glue**, allowing for quicker identification and resolution of model performance issues.
- **Enhanced team productivity by 40%** by mentoring junior developers in best practices for **AWS services** such as **SageMaker** and **CloudFormation**, focusing on building, deploying, and scaling machine learning models and ensuring best practices for MLOps (Machine Learning Operations).
- **Increased code stability and reliability by 25%** by refining automated testing protocols within CI/CD pipelines, particularly for **ML model validation and integration testing**, ensuring high accuracy and performance in model deployments.
- **Delivered a 20% cost reduction in operational expenditure** by optimizing backend architecture using **AWS Lambda** and **Step Functions**, specifically ensuring efficient scaling for **serverless ML models**, thereby reducing unnecessary resource consumption.

- **Facilitated a seamless shift to infrastructure as code (IaC)** with **AWS CloudFormation**, automating **model deployment and infrastructure changes**, reducing manual configuration errors by 40%, and ensuring reproducibility for **machine learning environments**.
- **Implemented real-time data synchronization across services** using **AWS EventBridge** and **Step Functions**, reducing latency in communication between ML services, and enabling faster real-time updates for **model inference**.
- **Spearheaded cross-functional collaboration to resolve critical production issues**, improving response times for critical incidents by 30%, while troubleshooting **ML model deployment issues** and enhancing performance through **AWS Lambda** and **Glue**.
- **Led the development of a cloud management dashboard using React**, improving real-time data visualization speed by 40%, enabling **dynamic monitoring of machine learning model performance** and providing insights to 100,000+ users.
- **Refactored a large-scale legacy application into a modern React architecture**, reducing technical debt by 30%, and improving the **front-end interface for displaying AI-driven insights** and predictions from deployed machine learning models.
- **Optimized frontend performance** of AWS services portal, reducing page load times by 50%, which directly enhanced **user engagement with machine learning-powered recommendations and insights**.
- **Refactored legacy monolithic applications to a microservices architecture** using **Node.js and AWS Lambda**, improving scalability for **AI and ML systems** by 50%, enabling rapid deployment of new machine learning models and reducing operational overhead.
- **Integrated multiple AWS services (S3, DynamoDB, RDS)** with **Node.js backend**, optimizing **data storage** and **retrieval for machine learning datasets**, reducing latency for **ML-based applications** by 30% and enabling faster model inference.
- **Implemented an event-driven architecture using Node.js and AWS SNS/SQS**, improving system responsiveness by 25%, enhancing the speed of **real-time predictions** and reducing processing errors for **high-traffic AI/ML applications**.

Tools: AWS (Glue, Lambda, SageMaker, CodePipeline), Python, Java, Node.js, CI/CD, Microservices, MLOps, ETL, React.js, Serverless, IaC (CloudFormation), Real-Time Data Sync, Event-Driven Architecture, Performance Optimization, Model Deployment.

Client: WIDES Labs, Los Angeles, CA (Remote) **June 2022 – Oct 2023**
Role: Python developer, ML

Responsibilities:

- **Designed and Developed Scalable Python Backend Solutions**
  Architected and implemented Python-based backend services for satellite communication systems, ensuring efficient data processing and integration with machine learning models for real-time performance optimizations.
- **Built and Deployed Machine Learning Models into Production**
  Developed and deployed ML models using **Python**, **PyTorch**, and **RLlib** to handle dynamic satellite network traffic. Reduced **latency by 18%** and **increased throughput by 20%** by integrating reinforcement learning-based algorithms into backend services.
- **Implemented RESTful APIs for ML Model Integration**
  Designed and built **RESTful APIs** in **Python** using **Flask** and **FastAPI** for seamless integration of machine learning models into satellite network systems. Enabled efficient communication between **backend services** and **frontend applications** (React) for real-time data updates.

- **Optimized Backend Performance Using AWS Cloud**
  Leveraged **AWS** services like **EC2**, **Lambda**, and **S3** to scale Python backend applications for high availability and fault tolerance. Optimized data storage and processing, reducing infrastructure costs by **20%** while improving system response time by **25%**.
- **Developed and Deployed AI-Powered Network Resource Allocation Systems**
  Created AI-driven backend algorithms for adaptive network resource allocation in **LEO satellite networks**, leveraging **RL** to dynamically adjust bandwidth and reduce congestion. Improved **network stability by 25%** and achieved **30% better load balancing**.
- **Collaborated on Full-Stack Development with React**
  Worked closely with the frontend team to implement **React** components for visualizing satellite network data. Integrated backend APIs with the React frontend, providing users with intuitive dashboards to monitor real-time network performance.
- **Created Cloud-Native ML Pipelines with AWS**
  Developed cloud-native ML pipelines on **AWS** to automate training and deployment of reinforcement learning models. Reduced model training time by **35%** through the use of **AWS S3** for data storage and **AWS SageMaker** for managed model training.
- **Optimized Satellite Data Processing with OpenCV & Python**
  Built and optimized **Python** image-processing pipelines using **OpenCV** for satellite data analysis. Enhanced data accuracy by **10%** through advanced image preprocessing and augmentation techniques, feeding cleaned data into ML models for network optimization.
- **Implemented Real-Time Data Processing and Model Updates**
  Integrated machine learning models into backend services to process **real-time satellite data** streams. Ensured models were continuously updated with fresh data through scheduled tasks using **AWS Lambda** and **Python**, improving model performance by **15%**.
- **Designed and Deployed CI/CD Pipelines for Python Backend**
  Implemented **CI/CD pipelines** using **GitLab CI** and **AWS CodePipeline**, automating deployment processes for Python backend applications and ML models. Reduced deployment times by **40%** and minimized errors during updates.
- **Enhanced Human-Computer Interaction (HCI) for Satellite Systems**
  Developed backend services to support **HCI enhancements** in satellite systems. Collaborated with frontend developers to create features that allowed users to interact dynamically with real-time satellite data, improving system accessibility.
- **Published Research and Contributed to Open Source**
  Published papers on integrating **machine learning** with satellite communication systems and **contributed to open-source ML projects**, focusing on the integration of ML models into Python backend services for scalable applications.

Tools: Python (Flask, FastAPI), Machine Learning, AWS (EC2, S3, Lambda), React.js, RESTful APIs, CI/CD (GitLab CI, AWS CodePipeline), Data Processing (OpenCV, NumPy), Docker, Kubernetes, SQL/NoSQL, Model Deployment.

Client: Tequed, India. **June 2019 – July 2021**
Role: Associate Software Developer

Responsibilities:

- Developed back-end components using **Python 3.7** on **Django 1.7 framework** and front-end using **HTML5, CSS/CSS3, JavaScrip**t to improve responsiveness and performance of **Web application**.
- Understanding of SQL queries and database management system such as **PostgreSQL**.

- Worked on application using python integrated IDEs like Eclipse, IntelliJ, Sublime Text and PyCharm.
- Installed data sources like **SQL-Server, Cassandra** and remote servers using the Docker containers as to provide the integrated testing environment for the **ETL applications**.
- Developed numerous new features and enhancements for the Web application using Python/Django, **HTML/CSS/JavaScript**, as well as automating our development environment using **Ansible**.
- Worked on optimizing and memory management of the **ETL applications** developed in Go-Lang and python and also reusing the existing code blocks for better performance.
- Provided ongoing maintenance, support and enhancements in existing systems and platforms, bug fixing and **tracking (JIRA)** and managing new application modules.
- Developed and executed White box test cases using **Python, Unitest/Pytest/Robot framework & PyCharm/Ride**.
- Created **GraphQL resolvers**, schemas, and data sources, so that can communicate with OpenShift APIs
- **Managed and optimized** large datasets using **Oracle database** tuning techniques and partitioning strategies, **resulting in a 20% reduction in storage costs**.
- **Automated routine financial reconciliation tasks** through **PL/SQL** scripts, **saving over 200 hours of manual work** monthly.
- **Created and deployed** high-performing batch jobs in **PL/SQL** to process financial transactions, **enhancing processing speed by 35%** for HSBC's daily batch processes.
- **Troubleshot and resolved** complex **Oracle database** issues, including performance bottlenecks and deadlocks, **reducing system downtime by 15%**.
- Developed dynamic web pages using **python Django Frameworks** and developed various templates for different customers and saved their user preferences.
- Developed views and templates with **Django's MVT (model, view, and template)** to create a user-friendly website.
- Microservice architecture development using Python and Docker on an Ubuntu Linux platform using HTTP/REST interfaces with deployment into a multi-node **Kubernetes** environment.
- Developed web applications and REST full web services and APIs using **Python, Django**.
- Developed tools using **Python, XML** to automate some of the manual tasks.
- Used **Python Flask** framework to build modular and maintainable applications.
- Used Celery with **Django** to complete asynchronous tasks, like huge report generation, sending mails to respective people.
- Wrote and executed various **PostgreSQL database** queries from python using **Psycopg2 connector**.
- Involved in **Agile methodology** and scrum process.
- Participated in day-to-day **meeting, status meeting, strong reporting** and effective communication with project manager and developers.

Environment: Python 3.7, Linux Suse, Slack , HTML5, OIDC, Ansible, Kubernetes, Django, Django Rest, Flask, React, Circle CI, Splunk, New Relic Server, GitHub, , PostgreSQL, GCP, EBS, PyCharm, Microsoft Visual Code, Linux, Shell Scripting, JIRA.